SAHADH FAZAL MOHAMED

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PROFESSIONAL SUMMARY

Software and AI Engineer who ships production systems rather than prototypes. My work is live: an iOS App Store application in daily operational use across 4 restaurant branches, and a RAG sales assistant that quotes exact prices to customers without a person in the loop. I design around failure modes, not just the happy path.

KEY ACHIEVEMENTS

- Shipped an iOS App Store app now central to daily operations across 4 branches; cut render latency by 20%.
- Designed a RAG pipeline with a 2 layer fallback that removed errors reaching customers during API outages.
- Kept a live routing API serving requests through cloud database outages via automated failover.
- Uncovered 11 defects across two developer implementations with an automated C++ unit test suite.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C++
Frameworks and Web: React, React Native, Node.js, Next.js, Flask, Figma (UI design)
AI and RAG: RAG pipeline design, vector databases (Supabase pgvector), embeddings and cosine similarity search, LLM integration (Google Gemini), agentic systems (ReAct), prompt engineering, PyTorch, TensorFlow
Systems and Cloud: Docker, Git, CI/CD (GitHub Actions), Azure, AWS, PostgreSQL, REST API design, failover patterns
Engineering: Data structures and algorithms, OOP, SOLID, system design, RBAC, Agile/Scrum, testing (pytest, Boost Unit Test Framework)

PROFESSIONAL EXPERIENCE

Cofounder and Lead Developer (Self employed), *Artim Labs* *Jan 2026 to Sep 2026*

- Led engineering for a UK software studio, architecting 6 live production systems for 7 client brands across retail, hospitality, automotive and travel in 7 months.
- Engineered TRUKBK's hallucination resistant AI sales assistant (Supabase pgvector, Google Gemini), grounding every answer in real product data rather than model recall.
- Built a 2 layer fallback into the retrieval pipeline so the assistant keeps answering through API rate limits and provider outages instead of surfacing errors to customers.
- Built TRUKBK's frontend configurator, cutting page load times by 1.2s.
- Owned delivery end to end across the client portfolio: requirements, architecture, deployment and post launch support.

Software Engineer (Freelance), *Arasan Travels* *Aug 2025 to Jan 2026*

- Built a full stack booking platform on a serverless pnpm monorepo with a type safe pipeline from database through to UI.
- Integrated Stripe payments with automated reconciliation against booking records, cutting admin overhead by 15%.
- Designed the schema and REST layer covering bookings, customers and payment records.

Software Developer (Contract), *PHAT Buns UK* *Apr 2024 to Dec 2025*

- Built PhatOps, an iOS operations app used daily by 20 staff across 4 branches, covering rostering, inventory and expense tracking.
- Designed the app's screens in Figma before building them, owning both the design and the engineering.
- Cut render latency by 20% through optimised components and tuned real time subscriptions.
- Shipped to the App Store and supported it in live use, iterating on feedback from branch managers.
- Sourced requirements from the branch staff using the app in live service, working from observed workflow rather than assumption.

PROJECTS

CycleNest API, *Java / JAX RS / Azure Cosmos DB* **[↗]** *Oct 2025*

- Designed a bike sharing routing API with an automated active failover pattern that switches to local in memory stores when the cloud database times out, then reconciles on recovery.

Car Logo Classifier, *Python / TensorFlow / NLTK* **[↗]** *Nov 2025*

- Built a VGG19 CNN for logo classification alongside a separate first order logic knowledge base (NLTK ResolutionProver): users teach the system new facts in plain English, and any fact that contradicts existing knowledge is proven false and rejected before it is stored.

GPS Journey Test Suite, *C++20 / Boost Unit Test Framework* **[↗]** *Mar to May 2025*

- Built an automated unit test suite validating a GPS journey detection algorithm against boundary conditions, invalid inputs and edge cases, uncovering 11 defects across two independent developer implementations, with the qmake and make build pipeline configured on Linux to automate compilation and test execution.

EDUCATION

Nottingham Trent University, *BSc (Hons) Computer Science* *Sep 2023 to Jun 2026*
Upper Second Class Honours (2:1). Relevant modules: AI and Machine Learning, Software Engineering. Student Representative, Sep 2023 to Sep 2024.

CIFA, *UK Financial Regulation and Professional Integrity (160 hours)* *Jun 2026 to Present*
Undertaken to support engineering work in regulated environments.

CERTIFICATIONS

IBM Certified iOS and Android Mobile App Developer | Microsoft Applied Skills: Enhance Agents with Autonomous Capabilities | Google AI Certification | Google Machine Learning Crash Course | AWS Cloud Foundations | Data Analysis with Python, freeCodeCamp